EV Technology Course — Module 15: Commercial EVs, Buses, Trucks and Fuel Cells

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Commercial vehicles drive many kilometres a day, so fuel savings add up quickly, but they also carry heavy loads, which needs very large batteries. Electric buses are already mainstream in India's cities; electric trucks are starting, helped by new incentives; and hydrogen fuel-cell vehicles are being tested for long-haul. This module explains the engineering and economics of each.

1. Learning objectives

- Describe the powertrain of electric buses, light and heavy trucks.
- Explain charging strategies: depot, opportunity (pantograph), megawatt charging, swapping.
- Explain how a fuel-cell electric vehicle (FCEV) works and compare it with a battery EV.
- Calculate daily energy, battery size and total cost of ownership (TCO).
- Understand Indian schemes and business models (GCC leasing) for e-buses and e-trucks.

2. Fundamentals

- **E-buses:** 9–12 m city buses with 250–450 kWh LFP batteries, 1.0–1.3 kWh/km, central motor or e-axle; depot charged overnight plus top-ups.
- **Light commercial vehicles (LCV, N1/N2):** small cargo vans and trucks with 20–100 kWh batteries; last-mile delivery is the earliest truck segment to electrify.
- **Heavy trucks (N3):** 300–900 kWh batteries, 1.1–1.6 kWh/km for 40-tonne combinations, high-power e-axles.
- **FCEV:** a hydrogen fuel cell generates electricity on board; a small battery buffers peaks; refuelling takes minutes.
- **TCO** (purchase + energy + maintenance + financing − residual) decides adoption, not purchase price alone.

3. Detailed technical explanation

E-bus architecture

- Low-floor body; battery on roof and/or rear; central PMSM with driveline or in-wheel/e-axle; electric air compressor (brakes, suspension), electric power steering pump, electric HVAC (large roof units), DC-DC for 24 V system.
- Charging: CCS2 plug at depot (120–240 kW), pantograph opportunity charging at terminals (300–600 kW).

E-truck architecture

- E-axle with two- to four-speed gearbox or multiple motors; batteries between frame rails and behind the cab; high voltage 600–1,000 V; electric power take-off (ePTO) for refrigeration, tippers.
- **Megawatt Charging System (MCS):** connector standard for up to ~3.75 MW; trucks charge in a driver's mandatory break (30–45 min).
- **Battery swapping** for trucks and buses is used in China and piloted in India for mining and port operations.

Fuel-cell electric vehicle

- **PEM fuel cell:** hydrogen at the anode splits into protons and electrons; protons cross a membrane; electrons flow through the external circuit (power); at the cathode they combine with oxygen to make water.
- Hydrogen stored at 350 bar (trucks, buses) or 700 bar (cars); in development, liquid hydrogen.
- Components: fuel-cell stack, air compressor, humidifier, hydrogen recirculation, DC-DC boost converter, buffer battery, cooling.
- Efficiency: fuel cell ~50–60%; well-to-wheel from renewable electricity roughly 25–35%, versus ~70–80% for battery EVs.

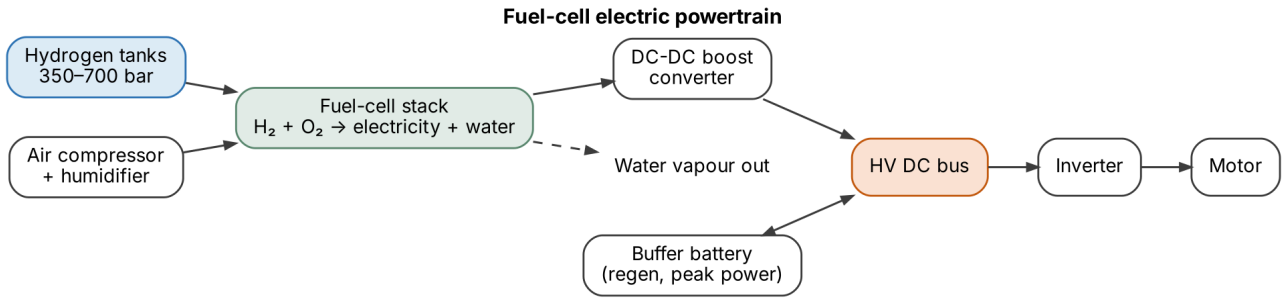
Comparison

	Battery e-truck	Fuel-cell truck	Diesel truck
Energy efficiency (well-to-wheel)	High (~75%)	Low (~25–35%)	Low (~20–25% tank-to-wheel)
Refuelling or charging	30–120 min	10–20 min	10–15 min
Payload penalty	Battery mass	Smaller	None
Infrastructure	Grid, chargers	Hydrogen production, storage, stations	Existing
Energy cost	Low	High today	Medium-high

India context

- **PM e-Bus Sewa** (2023) for 10,000 e-buses in smaller cities; **PM E-DRIVE** includes support for over 14,000 e-buses and, from July 2025, e-trucks.
- **E-truck incentive (July 2025):** ₹5,000 per kWh, limited by 10% of ex-factory price and a GVW-based cap (up to ₹9.6 lakh for 18.5–35 t trucks), requiring scrapping of an equivalent ICE truck.
- **GCC (gross cost contract):** operators own and run buses; state transport pays per km, reducing upfront burden.
- Key Indian players: Tata Motors, Olectra (with BYD technology), JBM, PMI, Switch Mobility (Ashok Leyland), EKA, Eicher (VECV); e-truck makers include Tata, IPLT, Blue Energy (LNG/EV), Volvo Eicher.
- **National Green Hydrogen Mission** supports hydrogen trials; pilots include NTPC hydrogen buses in Leh and Indian Oil fuel-cell buses.

4. Diagram: fuel-cell electric powertrain



5. Step-by-step: sizing an electric city bus

1. Collect route data: daily km, average speed, stops, gradient, AC hours, passengers.
2. Estimate consumption (kWh/km) from simulation or similar buses.
3. Daily energy = km × kWh/km; add 10–20% margin for ageing and weather.
4. Divide by usable SOC window (e.g. 80%) to get installed battery capacity.
5. Decide charging: overnight depot only, or depot + opportunity charging (smaller battery).
6. Check axle loads, passenger capacity, cost and TCO.
7. Validate on route trials before fleet order.

6. Engineering equations

Daily energy:

$$E_{day} = d_{day} \times c_{kWh/km}$$

Charging time:

$$t = \frac{E}{P_{charge}}$$

Cost per km:

$$\text{₹/km}_{diesel} = \frac{\text{₹/L}}{\text{km/L}}, \qquad \text{₹/km}_{EV} = c_{kWh/km} \times \text{₹/kWh}$$

Well-to-wheel efficiency (chain product):

$$\eta_{WTW} = \eta_1 \times \eta_2 \times \dots \times \eta_n$$

7. Worked examples

Example 1: bus energy. 250 km/day at 1.1 kWh/km = **275 kWh/day**. With 80% usable window and 15% margin, install about 395 kWh, or use a smaller battery with mid-day charging.

Example 2: e-bus vs diesel bus energy cost. Diesel at ₹90/L and 3 km/L = **₹30/km**. E-bus at 1.1 kWh/km and ₹8/kWh = **₹8.8/km**. At 80,000 km/year, saving ≈ **₹17 lakh/year** per bus.

Example 3: efficiency chain. Hydrogen route: electrolysis 70% × compression/transport 90% × fuel cell 50% × drive 90% = **28%**. Battery route: grid/charging 90% × battery 95% × drive 90% = **77%**. Battery EVs need roughly a third of the electricity per km.

Example 4: truck charging. A truck needing 500 kWh during a 30-minute break needs about **1 MW** of charging power, which is why MCS was developed.

8. Real-world examples

- **India e-buses:** thousands in Delhi, Mumbai (BEST), Bengaluru, Pune, Kolkata and Hyderabad under GCC contracts; mostly 9–12 m with LFP batteries.
- **Tesla Semi, Daimler eActros 600, Volvo FH Electric, BYD trucks:** long-haul battery trucks with 600–900 kWh packs.
- **China:** heavy e-truck sales grew rapidly from 2024, with battery-swap trucks in mining and ports.
- **Hyundai Xcient Fuel Cell** trucks in Switzerland and Korea; **Toyota** fuel-cell trucks; Indian pilots by NTPC, Indian Oil, Tata Motors and Ashok Leyland.

9. Manufacturing applications

- **Bus body building:** aluminium or steel frames, composite panels, roof-mounted battery structures; often low volume and high manual content.
- **Truck chassis:** ladder frame adapted for batteries; crash protection for side-mounted packs.
- **Fuel-cell stack production:** bipolar plates (stamped metal or graphite), membrane electrode assemblies (MEA), platinum catalyst; cost dominated by MEA and platinum at low volume.
- **Hydrogen tanks:** carbon-fibre wound Type IV tanks; carbon fibre is a big cost item.
- **Costing:** battery cost dominates e-bus and e-truck price; TCO models (₹/km) drive purchase decisions.

10. Common mistakes and misconceptions

- **"Hydrogen will replace batteries for all trucks."** Battery trucks are winning most short- and medium-haul uses; hydrogen may fit niche long-haul or high-utilisation uses.
- **"E-buses are expensive."** Purchase price is higher, but TCO is often lower at high kilometres.
- **"Bigger battery is always better for buses."** It reduces passenger capacity and raises cost; charging strategy matters.
- **"Fuel cells burn hydrogen."** They convert it electrochemically, not by combustion.

11. Practical case study: e-bus tender

Problem. A state transport undertaking wants 300 e-buses for routes of 180–260 km/day without raising fares.

Analysis. A GCC model at a fixed ₹/km price over 10–12 years; operator chooses 350 kWh battery with depot charging; depot needs ~8 MW grid connection; central subsidy reduces ₹/km.

Risks. Grid connection delays, battery degradation, operator payment delays, AC load in summer.

Lesson. E-bus success depends as much on depot power, contracts and finance as on vehicle technology.

12. Revision notes

- Buses: 250–450 kWh, ~1.1 kWh/km; trucks: 300–900 kWh, ~1.3 kWh/km.
- Charging: depot, opportunity (pantograph), MCS (MW-level), swapping.
- FCEV: H_2 + O_2 → electricity + water; WTW efficiency ~28% vs ~77% BEV.
- India: PM e-Bus Sewa, PM E-DRIVE e-buses and e-trucks (₹5,000/kWh, caps by GVW), GCC.

13. Glossary

Term	Meaning
GCC	Gross cost contract (pay-per-km bus model)
TCO	Total cost of ownership
MCS	Megawatt Charging System
PEM	Proton exchange membrane (fuel cell)
MEA	Membrane electrode assembly
ePTO	Electric power take-off
N1/N2/N3	Goods vehicle categories by weight

14. Knowledge check

Conceptual

1. Why are city buses an early success for electrification?
2. What is the main advantage of fuel cells for trucks?
3. Why is hydrogen less efficient than batteries?
4. What is a GCC model?
5. Why does the e-truck incentive require scrapping?

Numerical

1. A 12 m bus uses 1.2 kWh/km for 220 km/day. Daily energy?
2. A truck has 600 kWh installed with 90% usable at 1.4 kWh/km. Range?
3. A 300 kWh battery charged at 150 kW. Time (ignore taper)?

Application

1. Build a simple TCO comparison structure for e-bus vs diesel bus.
2. Which Indian truck segments would you electrify first and why?

15. Answer key

Conceptual

1. Fixed routes, depot charging, high daily kilometres, public funding, urban air-quality benefits.
2. Fast refuelling and lighter energy storage for long range.
3. Losses at electrolysis, compression and the fuel cell multiply.
4. The operator owns and runs buses; the authority pays a fixed amount per km.
5. To remove old polluting trucks and target the subsidy at replacing diesel.

Numerical

1. 220 × 1.2 = **264 kWh**.
2. 600 × 0.9 ÷ 1.4 = **386 km**.
3. 300 ÷ 150 = **2 hours**.

Application

1. Capex (vehicle, chargers, depot power) less subsidy; financing; energy cost per km; maintenance per km; battery replacement or warranty; insurance; driver and overheads (same); residual value; total ÷ lifetime km = ₹/km.
2. Urban last-mile LCVs and intra-city distribution, port and mining shuttles, cement and steel plant logistics, fixed-route short-haul: predictable routes, depot charging, high utilisation.

16. Further learning

- ICCT reports on e-buses and e-trucks in India.
- NITI Aayog and WRI India zero-emission truck studies.
- US DOE Hydrogen and Fuel Cell Technologies Office primers.
- CharIN documents on Megawatt Charging System.

Sources

- PM E-DRIVE incentive framework for electric trucks — EVreporter (MHI notification, 10 Jul 2025)

Next: Module 16, new and emerging EV technologies.